

# AMD Ryzen 7 7700 Desktop Processor

Best bang for your buck

**L**ooking at the scores of the AMD Ryzen 7 7700 at the end of our testing, we can make an outlandish claim that AMD held back the launch of the 65W processors simply because the 'X' series CPUs would not have sold alongside the 65W counterparts. Why? Because they're that good in terms of value proposition. Here's what we found out –

## SPECIFICATIONS

The AMD Ryzen 7 7700 is an 8-Core processor with multithreading much like all the other processors that AMD has within the Ryzen 7000 family. It's based on the Zen 4 architecture and is built for the new AMD AM5 socket. Compared to the Ryzen 9 processors which had 64 MB of L3 cache, the 7700 has to make do with half of that amount. L1 and L2, however, remain untouched on a per-core basis. This is not a different die than the previously launched Ryzen 7000 processors, so structurally there's not much different aside from the disabled cores.

So it goes without saying that even the iGPU is the same and consists of two AMD Radeon Graphics cores clocked at 400 MHz with a boost clock of 2200 MHz. The CPU boost clock has only been dropped by 100 MHz so that's not much of a hit. The base clock speed, however, has been reduced by 700 MHz down to 3.8 GHz. Now, that's quite a hit. Which should make you wonder whether you should really get the Ryzen 7 7700 when the 7700X is selling for just INR 1600 more? How about we look at the performance numbers to answer that.

## PERFORMANCE

We're going to compare the Ryzen



7 7700 with the 7700X since the two are closely priced. In Cinbench, the 7700 scores 18627 which is 6 per cent lesser than the 7700X. The single core score is also lower by about 4.4 per cent. In Blender, the score difference is about 3.2 per cent and in some programming workloads, the difference is just 1 per cent.

The story continues across the entire gamut of benchmarks that we ran with just five benchmarks showing a difference that's more than five percent. The average difference across all benchmarks between the two processors is 0.028 per cent. Which is to say that the two processors are practically the same.

So which processor should one get? The 7700X is just INR 1600 bucks more than the 7700 and you get about 700 MHz extra base clock. So, that's obviously a very meagre difference in cost for a supposed boost in performance. Honestly, we're going to side

with the 7700 here because of the simple fact that power consumption matters. INR 1600 might be small change when the processor budget is approximately INR 35K, however, when you consider the impact of the higher power consumption over time and the impact it has on your utility billing then the Ryzen 7 7700 trumps the 7700X in terms of value.

At idle, the 7700 clocks in at about 15-16 Watts whereas the 7700X hits about 23-24 Watts. And during load times, the difference is much higher as the 7700X ends up consuming between 130-140 Watts whereas the 7700 barely crosses 85 Watts. So, not only is there a decent gap during peak workloads but even with the processor idling, there's less power consumption. So we're saving on a lot of power without losing out on much performance. And that too while spending less money.

## VERDICT

The AMD Ryzen 7 7700 provides more value for money than most of the processors in the Ryzen 7000 family because it's not only providing the highest value for the money spent but it's

saving you more money over time thanks to the lower power consumption. We're yet to check out some of the competing processors from Intel to see if Team Blue has some hidden gems like the 7700 but for now, if you're going for the AM5 platform, then the Ryzen 7 7700 seems to be an absolute no brainer for getting the most out of your money without losing out on performance.

–Mithun Mohandas

## SPECIFICATIONS

Cores: 8 | Threads: 16 | Base clock: 3.8 GHz | Boost clock: 5.3 GHz | Process node: 5nm | L3 Cache: 32 MB | L2 Cache: 1 MB per core | TDP: 65 W.

## CONTACT

AMD | Phone: NA | Email: support.amd.com | Website: www.amd.com

